Case Study 1: Employee Payroll and Deduction Analysis

1. Problem Statement

Calculate an employee's gross salary from basic pay, allowances, overtime, and incentives. Apply deductions and print a formatted salary statement.

2. Objectives

Calculate gross salary and total deductions.

Use variables, arithmetic operators, and conditional statements.

Display a readable salary statement.

3. Assumptions

HRA is 20% of basic pay.

Travel allowance is entered by the user.

Overtime pay is entered as an amount.

Incentive is entered as an amount.

Deduction is 5% of gross salary when gross salary is at least ■25,000; otherwise it is 2%.

4. Algorithm

1. Start.

2. Read employee name, basic pay, travel allowance, overtime pay, and incentive.

3. Calculate HRA = 20% of basic pay.

4. Calculate gross salary by adding basic pay, HRA, travel allowance, overtime pay, and incentive.

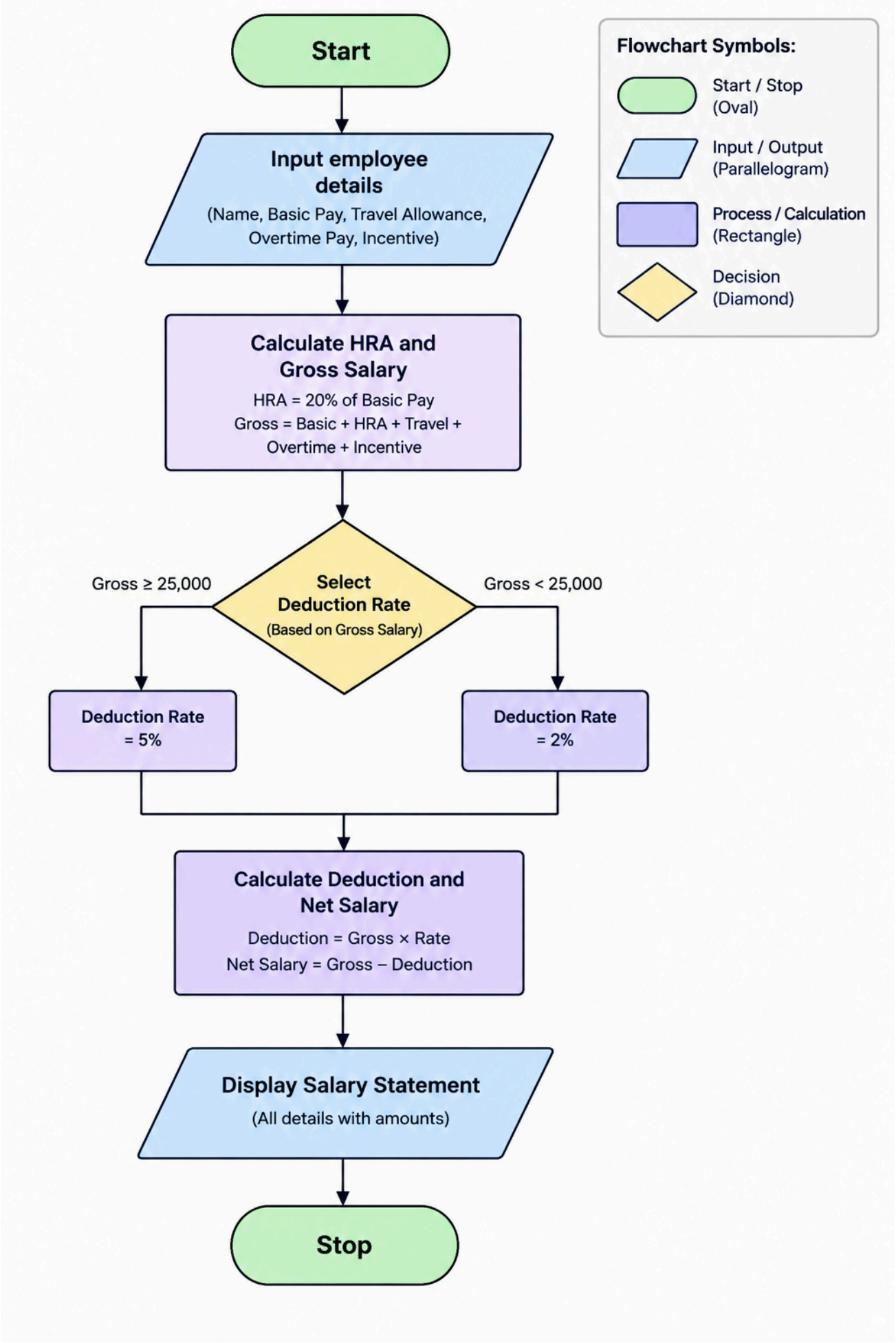
5. Choose the deduction rate using the gross salary condition.

6. Calculate deduction and net salary.

7. Display the salary statement.

8. Stop.

5. Flowchart :



6. Pseudocode

START
INPUT name, basic, travel, overtime, incentive
hra = basic * 0.20
gross = basic + hra + travel + overtime + incentive
IF gross >= 25000 THEN rate = 0.05 ELSE rate = 0.02
deduction = gross * rate
net = gross - deduction
DISPLAY salary statement

STOP

7. Complete Python Source Code

```
name = input("Enter employee name: ")
basic = float(input("Enter basic pay: Rs. "))
travel = float(input("Enter travel allowance: Rs. "))
overtime = float(input("Enter overtime pay: Rs. "))
incentive = float(input("Enter incentive: Rs. "))

hra = basic * 0.20
gross = basic + hra + travel + overtime + incentive

if gross >= 25000:
rate = 0.05
else:
rate = 0.02

deduction = gross * rate
net_salary = gross - deduction

print("
----- SALARY STATEMENT -----")
print("Employee:", name)
print(f"Basic pay: Rs. {basic:.2f}")
print(f"HRA: Rs. {hra:.2f}")
print(f"Travel allowance: Rs. {travel:.2f}")
print(f"Overtime pay: Rs. {overtime:.2f}")
print(f"Incentive: Rs. {incentive:.2f}")
print(f"Gross salary: Rs. {gross:.2f}")
print(f"Deduction ({rate * 100:.0f}%): Rs. {deduction:.2f}")
print(f"Net salary: Rs. {net_salary:.2f}")
```

8. Test Case and Expected Output

Inputs: name = Ravi,

basic = 20000,

travel = 1000,

overtime = 500,

incentive = 1000.

Gross salary = 20000 + 4000 + 1000 + 500 + 1000 = ■26,500.

Deduction = 5% of ■26,500 = ■1,325. Net salary = ■25,175.

9. Conclusion

The program calculates gross salary, applies a conditional deduction rate, and prints the employee’s net salary.